

Zookeeper 集群安装部署

简介

Zookeeper 是一个分布式的、开放源码的分布式应用程序协调服务, 是 Hadoop 和 Hbase 的重要组件, 它是一个为分布式应用提供一致性服务的软件, 提供的功能包括: 配置维护、域名服务、分布式同步、组服务等

安装

Zookeeper 是一款分布式的集群化软件, 可以在多台服务器部署, 并协同组成分布式集群一起工作

1. 【node1 操作】: 下载 Zookeeper 的安装包, 并解压

使用清华官方可用路径

wget https://archive.apache.org/dist/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz

```
[root@node1 ~]# wget https://archive.apache.org/dist/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz
--2024-04-25 17:45:00-- https://archive.apache.org/dist/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz
正在解析主机 archive.apache.org (archive.apache.org)... 55.168.204.189
正在连接 archive.apache.org (archive.apache.org)[65.168.204.189]:443... 已连接
已发出: HTTP 请求, 正在等待响应... 200 OK
长度: 1468463 (1.4M) [application/gzip]
正在保存为: "apache-zookeeper-3.8.4-bin.tar.gz"
apache-zookeeper-3.8.4-bin.tar.gz 100% [409.80K 13.9KB/s 剩余 11s 39s]
```

安装完成后, 把压缩包解压到 /export/server 目录下

```
[root@node1 ~]# ls
公共 模板 视频 图片 文档 下载 音乐 桌面 anaconda-ks.cfg apache-zookeeper-3.8.4-bin.tar.gz elasticsearch-8.17.3-x86_64.rpm hello.txt repo-backup
[root@node1 ~]# tar -zxvf apache-zookeeper-3.8.4-bin.tar.gz -C /export/server/
```

创建软链接

```
[root@node1 server]# ln -s /export/server/apache-zookeeper-3.8.4-bin /export/server/zookeeper
[root@node1 server]# ls
apache-tomcat-8.5.100 apache-zookeeper-3.8.4-bin jdk jdk1.8.0_491 tomcat8 zookeeper
[root@node1 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 6 root root 133 4月 25 17:57 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 41 4月 25 18:00 zookeeper -> /export/server/apache-zookeeper-3.8.4-bin
[root@node1 server]#
```

2. 【node1 操作】: 安装完成后, 进入 /export/server/zookeeper/conf/ 目录下,

将 zoo_sample.cfg 改名为 zoo.cfg, 并进入该文件, 修改参数

```
[root@node1 server]# cd ./zookeeper/
[root@node1 zookeeper]# ll
总用量 36
drwxr-xr-x. 2 xcxc xcxc 4096 2月 13 2024 bin
drwxr-xr-x. 2 xcxc xcxc 72 2月 13 2024 conf
drwxr-xr-x. 5 xcxc xcxc 4096 2月 13 2024 docs
drwxr-xr-x. 2 root root 4096 4月 25 17:57 lib
-rw-r--r--. 1 xcxc xcxc 11358 2月 13 2024 LICENSE.txt
-rw-r--r--. 1 xcxc xcxc 2084 2月 13 2024 NOTICE.txt
-rw-r--r--. 1 xcxc xcxc 2335 2月 13 2024 README.md
-rw-r--r--. 1 xcxc xcxc 3570 2月 13 2024 README_packaging.md
[root@node1 zookeeper]# cd ./conf/
[root@node1 conf]# ll
总用量 16
-rw-r--r--. 1 xcxc xcxc 535 2月 13 2024 configuration.xsl
-rw-r--r--. 1 xcxc xcxc 4559 2月 13 2024 logback.xml
-rw-r--r--. 1 xcxc xcxc 1183 2月 13 2024 zoo_sample.cfg
[root@node1 conf]# mv zoo_sample.cfg zoo.cfg
[root@node1 conf]# vim zoo.cfg
[root@node1 conf]#
```

修改之前

```
# The number of milliseconds of each tick
tickTime=2000
# The number of ticks that the initial
# synchronization phase can take
initLimit=10
# The number of ticks that can pass between
# sending a request and getting an acknowledgement
syncLimit=5
# the directory where the snapshot is stored.
# do not use /tmp for storage, /tmp here is just
# example sake.
dataDir=/tmp/zookeeper
# the port at which the clients will connect
clientPort=2181
# the maximum number of client connections.
# increase this if you need to handle more clients
#maxClientCnxns=60
#
# Be sure to read the maintenance section of the
# administrator guide before turning on autopurge.
"zoo.cfg" 37L, 1183B
```

修改之后

```
# The number of milliseconds of each tick
tickTime=2000
# The number of ticks that the initial
# synchronization phase can take
initLimit=5
# The number of ticks that can pass between
# sending a request and getting an acknowledgement
syncLimit=2
# the directory where the snapshot is stored.
# do not use /tmp for storage, /tmp here is just
# example sake.
dataDir=/export/server/zookeeper/data
# the port at which the clients will connect
clientPort=2181
# the maximum number of client connections.
# increase this if you need to handle more clients
#maxClientCnxns=60
#
```

添加内容如下

```
server.1=node1:2888:3888
server.2=node2:2888:3888
server.3=node3:2888:3888
```

```
~
~
~
```

3. 【node1 操作】:配置 myid

创建 zookeeper 的数据目录

```
[root@node1 conf]# vim zoo.cfg
[root@node1 conf]# mkdir /export/server/zookeeper/data
[root@node1 conf]# cd ..
```

创建一个文件 myid，写入内容 1，目的就是用来标识不同节点

```
[root@node1 data]# vim myid
[root@node1 data]# cat myid
1
[root@node1 data]#
```

4. 【node1 操作】：将 zookeeper 的安装包发送给 node2 和 node3

```
[root@node1 server]# scp -r apache-zookeeper-3.8.4-bin node2:`pwd`/
basic.css
chapter.gif
```

使用 scp 命令将安装包传输到 node2

node3 也是相同操作，检查是否发送成功

```
[root@node2 server]# cd /export/server/
[root@node2 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 7 root root 145 4月 25 18:22 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
[root@node2 server]#
```

```
[root@node3 ~]# cd /export/server/
[root@node3 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 7 root root 145 4月 25 18:22 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
[root@node3 server]#
```

node2 和 node3 已经收到安装包

5. 【node2 上操作】：创建软链接，把 myid 改为 2

```
[root@node2 server]# ln -s apache-zookeeper-3.8.4-bin /export/server/zookeeper
[root@node2 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 7 root root 145 4月 25 18:22 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node2 server]#
```

修改 myid

```
[root@node2 data]# cat myid
2
[root@node2 data]#
```

6. 【node3 上操作】：创建软链接，把 myid 改为 3

```
[root@node3 server]# ln -s apache-zookeeper-3.8.4-bin /export/server/zookeeper
[root@node3 server]# ll
总用量 8
drwxr-xr-x. 9 tomcat tomcat 4096 4月 22 21:06 apache-tomcat-8.5.100
drwxr-xr-x. 7 root root 145 4月 25 18:22 apache-zookeeper-3.8.4-bin
lrwxrwxrwx. 1 root root 28 4月 22 20:25 jdk -> /export/server/jdk1.8.0_491/
drwxr-xr-x. 8 root root 4096 4月 22 20:24 jdk1.8.0_491
lrwxrwxrwx. 1 tomcat tomcat 36 4月 22 21:09 tomcat8 -> /export/server/apache-tomcat-8.5.100
lrwxrwxrwx. 1 root root 26 4月 25 18:31 zookeeper -> apache-zookeeper-3.8.4-bin
[root@node3 server]#
```

修改 myid

```
[root@node3 data]# cat myid
3
[root@node3 data]#
```

7. 【在 node1,node2,node3 上分别执行】 启动 Zookeeper

```
[root@node1 zookeeper]# cd ./bin
[root@node1 bin]# ll
总用量 72
-rwxr-xr-x 1 xcxc xcxc 232 2月 13 2024 README.txt
-rwxr-xr-x 1 xcxc xcxc 1978 2月 13 2024 zkCleanup.sh
-rwxr-xr-x 1 xcxc xcxc 1115 2月 13 2024 zkCli.cmd
-rwxr-xr-x 1 xcxc xcxc 1576 2月 13 2024 zkCli.sh
-rwxr-xr-x 1 xcxc xcxc 1810 2月 13 2024 zkEnv.cmd
-rwxr-xr-x 1 xcxc xcxc 3613 2月 13 2024 zkEnv.sh
-rwxr-xr-x 1 xcxc xcxc 1243 2月 13 2024 zkServer.cmd
-rwxr-xr-x 1 xcxc xcxc 4559 2月 13 2024 zkServer-initialize.sh
-rwxr-xr-x 1 xcxc xcxc 11616 2月 13 2024 zkServer.sh
-rwxr-xr-x 1 xcxc xcxc 987 2月 13 2024 zkSnapshotComparer.cmd
-rwxr-xr-x 1 xcxc xcxc 1374 2月 13 2024 zkSnapshotComparer.sh
-rwxr-xr-x 1 xcxc xcxc 988 2月 13 2024 zkSnapshotToolkit.cmd
-rwxr-xr-x 1 xcxc xcxc 1377 2月 13 2024 zkSnapshotToolkit.sh
-rwxr-xr-x 1 xcxc xcxc 996 2月 13 2024 zkTxnLogToolkit.cmd
-rwxr-xr-x 1 xcxc xcxc 1385 2月 13 2024 zkTxnLogToolkit.sh
[root@node1 bin]#
```

进入/bin 文件目录下，找到一个 zkServer.sh 的程序，启动该程序

```
[root@node3 bin]# ./zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /export/server/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
[root@node3 bin]#
```

在三个节点都做相同操作，并输入 jps 指令

```
[root@node3 bin]# jps
279588 Jps
276209 QuorumPeerMain
1325160 -- process information unavailable
```

```
[root@node2 bin]# jps
276021 QuorumPeerMain
279634 Jps
1325160 -- process information unavailable
```

```
Starting zookeeper ... STARTED
[root@node3 bin]# jps
279588 Jps
276209 QuorumPeerMain
1325160 -- process information unavailable
```

都会出现一个同名进程 QuorumPeerMain，说明 zookeeper 启动成功

8. 【在 node1,node2,node3 上分别执行】：验证 Zookeeper

在/bin 目录下，找到一个 zkCli.sh 的程序，执行该程序

```
总用量 72
-rwxr-xr-x 1 root root 232 4月 25 18:22 README.txt
-rwxr-xr-x 1 root root 1978 4月 25 18:22 zkCleanup.sh
-rwxr-xr-x 1 root root 1115 4月 25 18:22 zkCli.cmd
-rwxr-xr-x 1 root root 1576 4月 25 18:22 zkCli.sh
-rwxr-xr-x 1 root root 1810 4月 25 18:22 zkEnv.cmd
-rwxr-xr-x 1 root root 3613 4月 25 18:22 zkEnv.sh
-rwxr-xr-x 1 root root 1243 4月 25 18:22 zkServer.cmd
-rwxr-xr-x 1 root root 4559 4月 25 18:22 zkServer-initialize.sh
-rwxr-xr-x 1 root root 11616 4月 25 18:22 zkServer.sh
-rwxr-xr-x 1 root root 987 4月 25 18:22 zkSnapshotComparer.cmd
-rwxr-xr-x 1 root root 1374 4月 25 18:22 zkSnapshotComparer.sh
-rwxr-xr-x 1 root root 988 4月 25 18:22 zkSnapshotToolkit.cmd
-rwxr-xr-x 1 root root 1377 4月 25 18:22 zkSnapshotToolkit.sh
-rwxr-xr-x 1 root root 996 4月 25 18:22 zkTxnLogToolkit.cmd
-rwxr-xr-x 1 root root 1385 4月 25 18:22 zkTxnLogToolkit.sh
```

执行程序

```
[root@node1 bin]# ./zkCli.sh
Connecting to localhost:2181
```

在最后出现的程序行输入 `ls /`，出现 zookeeper，说明该服务启动成功，记得三个节点都要尝试

```
WatchedEvent state:SyncConnected type:None path:null
[zk: localhost:2181(CONNECTED) 0] ls /
[zookeeper]
```

至此 Zookeeper 安装完成